Meteorological Data Dashboard - User Guide

Welcome to the **Interactive Spatial Data Visualisation** dashboard for Australian weather data. This comprehensive guide will help you navigate and utilize all features of the platform.

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Overview

The Meteorological Data Dashboard provides interactive visualization and analysis of Australian Bureau of Meteorology (BOM) weather data. The platform features:

- **377 weather stations** across Australia
- **Data from 2019 to August 2025**
- **5 key metrics**: Temperature, Rainfall, Humidity, Wind Speed, and Evapotranspiration
- **Real-time filtering** and aggregation
- **Statistical analysis** and trend detection
- **Export capabilities** for data and visualizations

Getting Started

First Visit

When you first visit the dashboard, you'll be greeted with a **Welcome Modal** that provides:

1. **Introduction** to the platform and its capabilities
2. **Interactive Tutorial** covering all major features (6 steps)
3. **Preference Settings** for saving filter configurations

Tip: Complete the tutorial to understand all available features. You can skip it and access the guide later via the "Guide" button in the header.

Interface Layout

The dashboard consists of:

- **Left Sidebar**: Filters and controls
- **Top Header**: Page title, theme toggle, export, and guide buttons
- **Main Content Area**: Map, charts, statistics, and insights
- **Mobile Support**: Responsive design with optimized layouts for smaller screens

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Dashboard Components

Interactive Map

The map is your primary interface for exploring weather stations.

Map Features

- **Station Markers**: Each weather station is represented by a marker
- **Clustering**: Stations automatically group when zoomed out
- **Heatmap Overlay**: Visualize metric intensity across regions
- **Interactive Selection**: Click stations to view detailed data

How to Use the Map

1. **Select a Station**:
    - Click on any station marker
    - The map centers on the selected station
    - Charts and statistics update to show station-specific data
    - The insights panel displays trend analysis for that station
2. **Deselect a Station**:
    - Click anywhere on the map background (not on a marker)
    - Returns to overview mode
    - Charts show aggregated data for the selected region/filters
3. **Navigation**:
    - **Zoom**: Use mouse wheel or +/- buttons
    - **Pan**: Click and drag the map
    - **Reset**: Click the home button to return to Australia-wide view
4. **Heatmap Toggle**:
    - Enable/disable via the sidebar
    - Shows color-coded intensity of the selected metric
    - Legend displays value ranges and colors
5. **Map Styles**:

- Choose from multiple base map styles in the sidebar:
  - Streets (default)
  - Satellite
  - Topographic
  - Light/Dark modes

Station Information Card

When you hover over a station marker, a popup displays:

- Station name
- Current metric value
- Location coordinates

Filter Sidebar

The sidebar provides comprehensive filtering and customization options.

Filter Options

1. **Time Period**:
   - **Start Date**: Select the beginning of your analysis period
   - **End Date**: Select the end of your analysis period
   - **Date Range Validation**: End date must be after start date
   - **Preset Ranges**: Quick selection for common periods
2. **Metric Selection**:
   - **Temperature** (°C): Daily average temperature
   - **Rainfall** (mm): Daily total precipitation
   - **Humidity** (%): Relative humidity percentage
   - **Wind Speed** (m/s): Average wind speed
   - **Evapotranspiration** (mm): Daily evapotranspiration
3. **Region Filter**:
   - Filter by Australian states and territories:
     - All Australia (default)
     - New South Wales (NSW)
     - Victoria (VIC)
     - Queensland (QLD)
     - South Australia (SA)
     - Western Australia (WA)
     - Tasmania (TAS)
     - Northern Territory (NT)
     - Australian Capital Territory (ACT)

Station Selection Priority ⚠

IMPORTANT: When you select a specific station on the map, the region filter automatically:

- Resets to "All Australia"
- Becomes disabled (grayed out)
- Shows a **red warning box** explaining the behavior

Why? This prevents conflicting filters. For example, analyzing a Sydney station (in NSW) with the region set to Victoria would produce no data.

How to re-enable region filtering:

- Click anywhere on the map to deselect the station
- The region dropdown will become active again
- You can then select a state/territory to analyze

Additional Sidebar Controls

- **Saved Views**: Save and load favorite filter combinations
- **Map Style Selector**: Change the base map appearance
- **Heatmap Toggle**: Enable/disable the heatmap overlay
- **Sidebar Collapse**: Minimize the sidebar for more viewing space

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Charts

The dashboard provides two main chart types for visualizing data.

Time Series Chart

Displays how metrics change over time.

Features:

- Line chart showing temporal trends
- Interactive tooltips with exact values
- Zoom and pan capabilities
- Download as PNG image

Aggregation Levels:

- **Daily**: Individual day values (default for short date ranges)
- **Weekly**: Average values per week
- **Monthly**: Average values per month
- **Yearly**: Average values per year

Note: Aggregation options automatically adjust based on your selected date range for optimal visualization.

How to Read:

- X-axis: Time period
- Y-axis: Metric value with appropriate units
- Hover over data points for exact values

Distribution Charts

Shows the statistical distribution of metric values.

Chart Types:

1. **Histogram**:
    - Displays frequency distribution
    - Shows how often values fall within ranges
    - Helps identify typical vs. extreme values
2. **Box Plot**:
    - Shows quartiles and outliers
    - Median line in the center
    - Box represents 25th-75th percentile
    - Whiskers show min/max (excluding outliers)
    - Outliers plotted as individual points

Use Cases:

- Identify data spread and variability
- Detect outliers and anomalies
- Understand typical value ranges
- Compare distributions across different time periods

Statistics Panel

Provides comprehensive statistical analysis of the filtered data.

Displayed Statistics

1. **Mean (Average)**:
    - The central tendency of the data
    - Sum of all values divided by count
2. **Median**:
    - The middle value when sorted
    - Less affected by outliers than mean
3. **Standard Deviation**:
    - Measure of data spread
    - Higher values indicate more variability
4. **Minimum Value**:
    - Lowest recorded value in the dataset

5. **Maximum Value**:
    - Highest recorded value in the dataset
6. **Data Points Count**:
    - Total number of observations in the filtered dataset

Correlation Analysis

When viewing data across multiple metrics:

- **Correlation Coefficients**: Shows relationships between metrics
- **Strength Indicators**: Weak, moderate, or strong correlations
- **Direction**: Positive or negative relationships

Example: Temperature and evapotranspiration often show positive correlation (higher temperature → higher evapotranspiration).

Insights Panel

Automatically analyzes your data to surface interesting patterns.

Trend Analysis

When a specific station is selected, the insights panel displays **Linear Regression Analysis**:

Components:

1. **Trend Direction**:
    - **Increasing** (shown in green): Values trending upward over time
    - **Decreasing** (shown in red): Values trending downward over time
    - **No significant trend** (shown in gray): Statistical analysis shows no clear pattern
2. **Statistical Significance**:
    - **p-value < 0.05**: Trend is statistically significant
    - **p-value ≥ 0.05**: Trend may be due to random variation
    - Lower p-values indicate higher confidence in the trend
3. **R^2 (R-squared) Value**:
    - Measures how well the trend line fits the data
    - Range: 0 to 1
    - Higher values (closer to 1) indicate better fit
    - Example: R^2 = 0.85 means 85% of variance is explained by the trend
4. **Slope**:
    - Rate of change per day
    - Units match the selected metric

- Example: Slope of 0.002°C/day = ~0.73°C increase per year

5. **Interpretation**:
   - Plain-language explanation of what the trend means
   - Automatically generated based on statistical analysis

Example Trend Analysis:

```
📈 Increasing Trend (Statistically Significant)

Temperature is increasing over the selected time period.

• Slope: 0.0019°C per day
• R²: 0.82 (strong fit)
• p-value: 0.0003 (highly significant)
• Data points: 1,825

Interpretation: Temperature shows a statistically significant increasing
trend of approximately 0.69°C per year at this station.
```

Anomaly Detection

Highlights:

- Identifies unusually high or low values
- Based on statistical thresholds (typically 2+ standard deviations)
- Displays dates and values of notable events

Use Cases:

- Detect extreme weather events
- Identify data quality issues
- Investigate unusual patterns

Key Features

1. Saved Views

Purpose: Save frequently used filter combinations for quick access.

How to Use:

1. Configure your desired filters (date range, metric, region, station)
2. Click "Save Current View" in the sidebar
3. Enter a descriptive name (e.g., "Sydney Summer 2024")
4. View appears in the "Saved Views" list
5. Click any saved view to instantly restore those filters

Benefits:

- Save time on repetitive analysis
- Track specific stations over time
- Compare seasonal patterns
- Share filter configurations via exported settings

2. Data Export

CSV Export:

1. Click the "Export" button in the header
2. Select which metrics to include
3. Download filtered data as CSV
4. Opens in Excel, Python, R, or other analysis tools

Export Contents:

- Date/time stamps
- Station information
- Selected metric values
- Filtered according to current settings

3. Chart Export

Image Export:

1. Navigate to the desired chart view
2. Click the download icon on the chart
3. Saves as PNG image file
4. Suitable for reports, presentations, and documentation

4. Theme Toggle

Light Mode:

- Default high-contrast theme
- Better for bright environments
- Easier on some screens

Dark Mode:

- Reduces eye strain in low-light conditions
- Saves battery on OLED screens
- Modern aesthetic

Toggle: Click "Light Mode" or "Dark Mode" button in the header

5. Responsive Design

The dashboard automatically adapts to different screen sizes:

- **Desktop** (1920px+): Full layout with all panels visible
- **Laptop** (1366px+): Optimized grid layout
- **Tablet** (768px+): Stacked layout with touch controls
- **Mobile** (< 768px): Single-column layout with simplified navigation

Mobile Features:

- Collapsible sidebar
- Touch-friendly controls
- Optimized chart sizes
- Screen size warning for very small devices

Advanced Usage

URL Parameters

The dashboard supports URL parameters for sharing specific views:

Examples:

```
?metric=temperature&start_date=2024-01-01&end_date=2024-12-31
?state=NSW&metric=rainfall
?station_id=066062&metric=humidity
```

Supported Parameters:

- `metric`: temperature, rainfall, humidity, wind_speed, evapotranspiration
- `start_date`: YYYY-MM-DD format
- `end_date`: YYYY-MM-DD format
- `state`: NSW, VIC, QLD, SA, WA, TAS, NT, ACT
- `station_id`: BOM station ID

Keyboard Shortcuts

- **Esc**: Close open modals (Welcome, Export, Guide)
- **Tab**: Navigate between interactive elements
- **Enter**: Confirm selections in modals

Performance Optimization

Tips for Smooth Performance:

1. **Limit Date Ranges**: Shorter periods load faster
2. **Use Aggregation**: Weekly/monthly views reduce data points
3. **Single Station Analysis**: Faster than state-wide analysis
4. **Disable Heatmap**: Improves map rendering speed
5. **Clear Browser Cache**: If experiencing slowdowns

Data Refresh

- Data is cached for 5 minutes to improve performance
- Changing filters triggers new API requests
- Large queries (multiple years, all stations) may take longer

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Troubleshooting

Common Issues

No Data Displayed

Possible Causes:

1. **Invalid Date Range**: End date before start date
2. **No Stations in Region**: Selected state may have limited coverage
3. **Station/Region Conflict**: Station selected outside filtered region
4. **Network Error**: Check internet connection

Solutions:

- Reset filters to defaults
- Try a different date range
- Deselect station to re-enable region filter
- Refresh the page

Charts Not Loading

Possible Causes:

1. **Browser Compatibility**: Older browsers may not support all features
2. **JavaScript Disabled**: Charts require JavaScript
3. **Ad Blockers**: May interfere with chart rendering

Solutions:

- Use modern browsers (Chrome, Firefox, Safari, Edge)
- Enable JavaScript
- Disable ad blockers for this site
- Clear browser cache

Slow Performance

Possible Causes:

1. **Large Dataset**: Querying many years of data
2. **Too Many Stations**: Heatmap with all stations can be slow
3. **Limited Resources**: Device has low memory

Solutions:

- Reduce date range
- Filter by single state/region
- Disable heatmap overlay
- Use data aggregation (monthly instead of daily)
- Close other browser tabs

Export Fails

Possible Causes:

1. **File Size Too Large**: Exporting too much data
2. **Popup Blocker**: Preventing download
3. **Insufficient Permissions**: Browser blocking file downloads

Solutions:

- Reduce date range before exporting
- Allow popups for this site
- Check browser download settings
- Try different export format

Browser Support

Recommended Browsers:

- ☑ Google Chrome (latest)
- ☑ Mozilla Firefox (latest)
- ☑ Microsoft Edge (latest)
- ☑ Safari (latest)

Not Supported:

- ❌ Internet Explorer
- ❌ Browsers with JavaScript disabled

Getting Help

If you encounter issues not covered here:

1. **Check Browser Console**: Press F12 to view error messages
2. **Try Incognito/Private Mode**: Rules out extension conflicts
3. **Contact Support**: Provide error messages and steps to reproduce

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Technical Details

Data Source

- **Provider**: Australian Bureau of Meteorology (BOM)
- **Stations**: 377 active weather stations
- **Coverage**: Australia-wide

- **Time Period**: 2019 - August 2025
- **Update Frequency**: Data imported from historical BOM records

Metrics Definitions

1. **Temperature (°C)**:
    - Daily average temperature
    - Calculated from hourly observations
    - Precision: 0.1°C
2. **Rainfall (mm)**:
    - Daily total precipitation
    - Measured in millimeters
    - Includes rain, drizzle, and melted snow
3. **Humidity (%)**:
    - Relative humidity percentage
    - Average of 9am and 3pm observations
    - Range: 0-100%
4. **Wind Speed (m/s)**:
    - Average wind speed
    - Measured in meters per second
    - 10-minute average
5. **Evapotranspiration (mm)**:
    - Daily evapotranspiration
    - Calculated using meteorological data
    - Represents water loss from soil and plants

Statistical Methods

Trend Analysis:

- Method: Linear Regression (Ordinary Least Squares)
- Library: scipy.stats.linregress
- Significance Level: $p < 0.05$ (95% confidence)

Anomaly Detection:

- Threshold: ±2 standard deviations from mean
- Method: Z-score analysis

Correlation Analysis:

- Method: Pearson correlation coefficient
- Range: -1 (perfect negative) to +1 (perfect positive)

Technology Stack

Frontend:

- React 18
- Vite (build tool)
- Tailwind CSS (styling)
- Leaflet (maps)
- Chart.js (charts)
- React Query (data fetching)

Backend:

- Python 3.11+
- Flask (web framework)
- PostgreSQL (database)
- SQLAlchemy (ORM)
- NumPy/SciPy (statistical analysis)

API Rate Limiting

To ensure fair usage and system stability:

- **Standard Endpoints**: 60 requests per minute
- **Export Endpoint**: 10 requests per minute
- **Trend Analysis**: 20 requests per minute

Exceeding limits returns HTTP 429 (Too Many Requests).

Data Privacy

- **No User Tracking**: Dashboard does not collect personal information
- **Local Storage**: Preferences saved locally in your browser
- **No Cookies**: Session management is client-side only
- **Anonymous Usage**: All weather data is public domain

Accessibility

The dashboard follows WCAG 2.1 Level AA guidelines:

- **Keyboard Navigation**: Full support for keyboard-only users
- **Screen Readers**: ARIA labels and semantic HTML
- **Color Contrast**: Meets minimum contrast ratios
- **Focus Indicators**: Clear visual focus states
- **Responsive Text**: Scales with browser zoom

Appendix

Glossary

- **Aggregation**: Combining multiple data points (e.g., daily → monthly averages)
- **Anomaly**: A value significantly different from the norm
- **Correlation**: Statistical relationship between two variables
- **Heatmap**: Color-coded visualization of spatial data
- **Linear Regression**: Statistical method to identify trends
- **p-value**: Probability that results occurred by chance
- **R^2**: Measure of how well a trend fits the data
- **Standard Deviation**: Measure of data variability

Useful Links

- **BOM Website**: http://www.bom.gov.au
- **Station Metadata**: Available in CSV export

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For questions or feedback, please contact the development team.